

Vascular Model Repository

Specifications Document



0266_H_CORO_CAD

Legacy Name: ASOCA-18

Model added: 22 Sep 2025

Species	Human
Anatomy	Coronary
Disease	Coronary Artery Disease
Procedure	None

Clinical Significance and Background

Coronary

To function, like all other tissues in the body, the heart muscle needs oxygen-rich blood to be delivered and oxygen-depleted blood to be carried away. Coronary arteries wrap around the outside of the heart, supplying blood to the heart muscle. The two main coronary arteries are the left main and right coronary arteries. Small branches of the coronary artery dive into the heart muscle to bring it blood.

The left main coronary artery (LCMA) supplies blood to the left side of the heart muscle, the left ventricle and left atrium. The left main coronary divides into two branches, the left anterior descending artery and the circumflex artery. The left anterior descending artery supplies blood to the front of the left side of the heart, and the circumflex artery encircles the heart muscle supplies blood to the outer side and back of the heart.

The right coronary artery (RCA) supplies blood to the right ventricle, the right atrium, and the sinoatrial (SA) and atrioventricular (AV) nodes, which regulate the heart rhythm. The right coronary artery divides into smaller branches, such as the right posterior descending artery and the acute marginal artery. With the left anterior descending artery, the right coronary artery helps supply blood to the middle, or septum, of the heart.

Clinical Data

General Patient Data

Age (yrs)	50
Sex	Male

Notes

A 50-year-old male patient with atherosclerotic coronary artery disease diagnosed by CTCA. Access to the raw CTCA images and expert annotations can be requested at <https://reshare.ukdataservice.ac.uk/855916/>. See below for information on the image data.

Image Modality: CTCA

Publications

See the following publications which include the featured model for more details:

Gharlegghi, R., Adikari, D., Ellenberger, K. et al. Annotated computed tomography coronary angiogram images and associated data of normal and diseased arteries. Sci Data 10, 128 (2023). <https://doi.org/10.1038/s41597-023-02016-2>

Gharlegghi, R., Adikari, D., Ellenberger, K. et al. Automated segmentation of normal and diseased coronary arteries - The ASOCA challenge. Comput Med Imaging Graph 97, 102049 (2022). <https://doi.org/10.1016/j.compmedimag.2022.102049>

Zhang, M., Gharlegghi, R., Shen, C. et al. A new understanding of coronary curvature and haemodynamic impact on the course of plaque onset and progression. R Soc Open Sci 11, 241267 (2024). <https://doi.org/10.1098/rsos.241267>

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AND/OR

N.M. Wilson, A.K. Ortiz, and A.B. Johnson, "The Vascular Model Repository: A Public Resource of Medical Imaging Data and Blood Flow Simulation Results," J. Med. Devices 7(4), 040923 (Dec 05, 2013) doi:10.1115/1.4025983.

AND/OR

Reference the official website for this data: www.vascularmodel.com

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